A Taxonomy of Knowledge Graph-Enhanced Dialogue State Tracking and ConvoTree: A Neural Consolidation Engine

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Abstract

We present a systematic taxonomy of Knowledge Graph-enhanced Dialogue State Tracking (DST) methods, organizing approaches by four key dimensions: graph construction strategy, integration with language models, scope and domain coverage, and computational efficiency. Unlike previous chronological surveys, our taxonomy reveals fundamental design patterns and identifies convergence toward hybrid architectures that combine graph reasoning with large language models. We demonstrate these principles through ConvoTree, an open-source proof-of-concept system that explores neural consolidation for conversation memory management. ConvoTree implements dynamic graph construction with LLM-powered reasoning to discover relationships and consolidate knowledge across dialogue sessions. The system provides a command-line interface for experimenting with knowledge graph-based conversation persistence and demonstrates the potential for addressing context limitations in extended dialogues. Our taxonomic analysis and prototype implementation provide a foundation for future research in graph-enhanced dialogue systems.

1 Introduction

Task-oriented dialogue systems must track user goals across conversational turns while managing growing context windows. Modern day models are able to take user queries in unstructured natural language format and generalize to unseen computations through multiple techniques (Dong et al., 2023; Wei et al., 2023; Bai et al., 2022).

End to end approaches sample tokens autoregresively by attending over the tokens from the previous dialogue turns with transformers. The model is run until the end of turn token is generated and the user is prompted for further input. Generated tokens depend on all previous tokens in the context and learned parameters in transformer layers. As context length increases, computational load grows and retrieval quality diminishes (Nelson et al., 2024; Hong et al., 2025). While strict care is taken to enrich and contain as much as possible data in the pretraining and the RLHF stages it is not possible to keep the model constantly updated and have wide task coverage for logistics and economics reason. Task specific procedures have to be included in the context through search, retrieval and augmentation. Dialogue state tracking addresses this by maintaining structured representations of user goals, while knowledge graphs can enhance this tracking through explicit relational modeling.

In task-oriented dialogue research, the domain designates the topical sphere—such as restaurant or hotel—within which the system must act. Each domain is governed by an ontology (sometimes called a schema), that is, a catalogue of slots (attributes) and their admissible slot values. A slot corresponds to a particular attribute—for example, price-range—while a slot value is its concrete instantiation, as when the user specifies "cheap." After every conversational turn, the system maintains a belief state: a probabilistic hypothesis over the possible slot–value pairs that represent what it currently thinks the user wants, which evolves as the dialogue unfolds. Utterances themselves are characterised by dialogue acts such as inform, request, or confirm; these acts, together with recognised intents like book-table, serve as abstractions of communicative function. A turn (or frame) comprises one user utterance followed by one system response, and the ordered sequence of turns constitutes the dialogue history.

Knowledge Graphs are a subtype of directed graphs, where there is a label attached to the edge denoting the semantic relationship between the source, destination vertices. Each vertex is populated by either an entity or a property of such entities. They can be used to structure a dataset into a graph format enabling storage, visualization and multi-hop reasoning over the knowledge base. Multi-hop reasoning leverages the connectivity of multiple nodes to enable enriched context with semantically relevant information to be pooled into the context from neighboring vertices.

State representation of the dialogue with task defined slots are structured to make interfacing with external knowledge and rule bases much easier. These tasks might require business logic to be adhered to, the context to be pieced over multi turns and missing slots to be prompted from the user. Knowledge graphs can integrate with task oriented dialogue systems by providing an interface for external information, however the semantically related graph representation also successfully models the dynamic and structured dialogue state.

This paper organizes knowledge graph-enhanced DST approaches into a systematic taxonomy and presents ConvoTree, an implementation that combines graph construction with memory consolidation techniques. Related Work section will introduce the seminal papers of the fields and put the relevant terms into their context while analyzing the anthology of the research. Methodology section is to introduce Knowledge Graphs for Dialogue State Tracking through application, evaluation, methods. In the Discussion section we will weigh the benefits, downsides of the approach and review the case for wider application of these systems to task oriented dialogue systems.

2 Related work

Dialogue State Tracking (DST) is the component of a task-oriented dialogue system that keeps a structured snapshot of user goals, constraints, and the system's own commitments after each turn. Early rule-based trackers relied on handcrafted finite-state machines and explicit slot-filling logic, which proved brittle and expensive to port across languages and tasks (Paek and Pieraccini, 2008).

The limitations of manual design motivated the move to statistical models: treating DST as a partially observable Markov decision process enabled Bayesian belief tracking and reinforcement-learning policies (Young et al., 2013). The Dialogue State Tracking Challenge (DSTC) series, launched in 2013, established shared evaluation frameworks (Williams et al., 2016). Early challenges progressed from single-domain telephone tasks (Henderson et al., 2013) to multi-domain restaurant booking (Henderson et al., 2014a), culminating in schema-guided tracking that requires reasoning over unseen APIs (Kim et al., 2019a).

Neural approaches evolved from discriminative models mapping dialogue histories to slot distributions (Henderson et al., 2014b) to generative pointer-networks enabling zero-shot transfer (Wu et al., 2019). Modern graph-enhanced architectures explicitly model inter-slot and external knowledge relations, motivating systematic analysis of these design patterns (Chen et al., 2020; Lin et al., 2021a; Li et al., 2025).

Knowledge Graphs structure information as entity-relation-entity triples, originating with WordNet (Miller, 1995) and scaling to web-scale resources like Freebase and DBpedia (Bollacker et al., 2008; Lehmann et al., 2015). Representation learning methods embed these graphs for reasoning tasks (Bordes et al., 2013; Sun et al., 2019), while industrial adoption demonstrates their value for search and recommendation systems (Singhal, 2012). The structured nature of knowledge graphs provides natural frameworks for modeling dialogue state relationships and external domain knowledge, creating opportunities for enhanced state tracking through explicit relational reasoning.

3 Methods: A Taxonomy of Knowledge Graph-Enhanced DST

Knowledge Graph-enhanced dialogue state tracking (DST) has evolved from simple static schemas to sophisticated dynamic architectures that adapt to conversation flow. Rather than presenting approaches chronologically, we propose a systematic taxonomy that organizes methods by their fundamental design principles. This classification reveals clear patterns in how graph structures address different DST challenges and guides future research directions.

3.1 Graph Construction Strategy

The first dimension of our taxonomy distinguishes how graph structures are created and maintained throughout dialogue processing.

3.1.1 Static Schema Graphs

Static approaches construct fixed graph structures based on domain ontologies, where edges encode predetermined relationships between slots, domains, or values. The Schema-guided State Tracker (SST) (Chen et al., 2020) exemplifies this approach, using human-defined edges to connect related slots within and across domains. Each slot becomes a graph node, with Graph Attention Networks propagating information along edges that represent domain membership or semantic compatibility. This allows slots like “restaurant-name” and “restaurant-food” to share information when connected by ontology-defined relationships. SST achieved 51–55% JGA on MultiWOZ 2.0–2.1, demonstrating that even fixed graph structures provide valuable inductive bias for multi-domain tracking.

3.1.2 Dynamic Turn-Level Graphs

Dynamic approaches construct or modify graph structures at each dialogue turn, adapting to conversation-specific patterns. DSTQA (Zhou and Small, 2019a) maintains an evolving schema graph where edges represent relations discovered during training and updated as conversations unfold. This enables cross-domain dependency modeling, such as linking “taxi-destination” to previously mentioned “restaurant-name.” More recent work like DSGFNet (Feng et al., 2022) combines static ontology edges with dynamic co-occurrence edges that activate when slots share evidence in the current turn, achieving more robust tracking in later dialogue turns.

3.1.3 Hierarchical Multi-Level Graphs

Advanced systems construct multiple interconnected graph structures operating at different granularities. HS2DG-DST (Gong et al., 2025) builds dual graphs per turn: a dialogue graph linking tokens and slots for coreference reasoning, and a value graph modeling domain–slot hierarchies. HFSG-DST (Liao et al., 2025) extends this concept by inserting syntactic-dependency and constituency nodes beneath each slot, creating hierarchical fine-grained state-aware graphs where token-level evidence influences slot predictions through message passing over richer structures.

3.2 Integration with Language Models

The second dimension categorizes how graph structures interface with neural language understanding components.

3.2.1 Graph-Augmented Transformers

These approaches integrate graph neural networks with transformer architectures, typically using graph attention layers to process structural relationships before or alongside standard self-attention. The GPT2-GAT model (Lin et al., 2021b) exemplifies this strategy, treating domain-slot pairs as graph nodes and applying GAT layers for inter-slot information exchange before GPT-2 decoding. This hybrid design allows structured schema knowledge to compensate for data scarcity in generative DST models.

3.2.2 Graph-Guided Generation and Reranking

More recent hybrid approaches use large language models for candidate generation while employing graph structures for verification and reranking. Noetic-Graph DST (Li et al., 2025) combines turn-level "noetic" graphs—explicitly representing slot value entailment, contradiction, or uncertainty—with T5 backbone generation followed by graph-feature-based reranking. This design achieves 78% JGA on MultiWOZ 2.4 by leveraging both the generative power of large models and the structural reasoning of graphs.

3.2.3 Learned Relational Mechanisms

Some approaches learn to capture graph-like relationships without explicit structural modeling. DST-STAR (Ye et al., 2021b) introduced slot self-attention mechanisms that automatically discover correlations among slots from data, achieving competitive performance (73.6% JGA on MultiWOZ 2.4) through purely learned relational modeling rather than predefined graph structures.

3.3 Scope and Domain Coverage

The third dimension distinguishes methods by their target application scenarios and coverage breadth.

3.3.1 Single-Domain Specialists

Early work focused on single-domain applications with relatively simple ontologies. GraphDialog (Yang et al., 2020) integrated dialogue dependency graphs with knowledge base graphs for restaurant booking, demonstrating that even within one domain, graph structures capture semantics that list-based approaches overlook through multi-hop reasoning over entity relationships.

3.3.2 Multi-Domain Generalizers

Most contemporary research targets multi-domain scenarios where dialogues span multiple service domains. These systems must handle inter-domain dependencies and slot relationships across different ontological structures. GCDST (Wu et al., 2020) constructs state graphs linking slots across domains when they share values or coreferential expressions, employing copy mechanisms to maintain consistency in complex multi-domain conversations.

3.3.3 Cross-Lingual and Multimodal Extensions

Advanced systems extend graph-based DST to multilingual and multimodal settings. XQA-DST (Zhou et al., 2022) uses domain-slot question templates as language-agnostic schema structures, achieving strong zero-shot transfer from English to German (66.2% JGA) and Italian (75.7% JGA). Multimodal approaches like the Video-Dialogue Transformer Network (Le et al., 2022) create latent graphs connecting visual object features with dialogue tokens for video-grounded conversations.

3.4 Computational Efficiency and Practical Deployment

The fourth dimension addresses the practical considerations of computational cost and deployment requirements.

3.4.1 Lightweight Graph Modules

Some approaches prioritize computational efficiency through streamlined graph architectures. ECDG-DST (Zhu and Xu, 2025) learns efficient context encoders gated by lightweight domain-guidance graphs, achieving 76% JGA while adding only 30ms latency per turn, making it suitable for production deployment.

3.4.2 Scalable Message Passing

Advanced systems address the $O(|S|^2)$ complexity of dense message passing through sparse or hierarchical formulations. These approaches become crucial when slot spaces expand to thousands of attributes in open-domain applications.

3.5 Synthesis and Future Directions

This taxonomy reveals several key insights about the evolution of graph-enhanced DST:

Convergence toward Hybrid Architectures: The most successful recent approaches combine large language model generation with graph-based verification, suggesting that future progress lies in tighter integration rather than pure graph-based or pure neural approaches.

Hierarchical Structure Benefits: Systems that operate at multiple granularities (token, slot, domain levels) consistently outperform flat representations, indicating the importance of multi-level graph modeling.

Dynamic Adaptation Advantages: Methods that adapt graph structures to conversation-specific patterns show superior performance over static schemas, particularly for long-range dependency tracking.

The taxonomy also identifies promising research directions: continual graph expansion for lifelong learning, cross-modal graph alignment for unified multimodal understanding, and efficient sparse graph formulations for web-scale deployment.

4 Discussion

The empirical evidence surveyed so far shows that graph-based Dialogue State Tracking (DST) consistently outperforms non-graph baselines across a variety of benchmarks. A core reason is that graphs inject *relational inductive bias* – encouraging the model to propagate information along edges that represent domain, slot, or value correlations. This section deepens that discussion by (i) formalising the evaluation metrics most commonly used to quantify these gains, (ii) reflecting on the comparative behaviour of graph versus non-graph architectures, and (iii) outlining future research trajectories.

4.1 Evaluation Benchmarks and Metrics

Evaluating DST models, including KG-based ones, requires standard benchmarks and metrics to ensure fair comparison. The MultiWOZ family of datasets (Budzianowski et al., 2018) is the most widely used evaluation benchmark for multi-domain DST. MultiWOZ 2.4 (Ye et al., 2021a) is the latest version with cleaned validation and test sets, enabling a more reliable evaluation. Typically, the primary metric is Joint Goal Accuracy (JGA) – the proportion of dialogue turns for which the model's predicted state exactly matches the ground truth for all slots. As JGA is strict, it is often complemented by Slot Accuracy. Beyond MultiWOZ, the Schema-Guided Dialogue (SGD) dataset (Rastogi et al., 2020) tests zero-shot generalization with additional metrics such as Active Intent Accuracy and Requested Slots F1. Graph-based approaches that leverage schema knowledge tend to excel on SGD because they better transfer to unseen domains.

4.2 Formal definitions of evaluation metrics

Let a dialogue contain T turns, and let $\mathcal{S}$ denote the ontology set of slots. For turn t, let the ground-truth belief state be $s_t = \{(s, v_{t,s}) : s \in \mathcal{S}\}$ and the model's prediction be $\hat{s}_t$.

Joint Goal Accuracy (JGA).

$$\mathrm{JGA} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{1}\big[\hat{s}_t = s_t\big],$$

where the indicator $\mathbf{1}[\cdot]$ equals 1 iff every slot–value pair is correct at turn t. As noted earlier, JGA is stringent: a single slot mismatch yields zero for that turn.

Slot Accuracy (SA).

$$\mathrm{SA} = \frac{1}{T\,|\mathcal{S}|}\sum_{t=1}^{T}\sum_{s\in\mathcal{S}}\mathbf{1}\big[\hat{v}_{t,s} = v_{t,s}\big].$$

SA gives partial credit, revealing that modern trackers achieve 96–98% accuracy per slot on MultiWOZ 2.4 yet still fall short in JGA because errors concentrate on different slots across turns.

Requested Slots F1 (RS-F1). For each turn, define the ground-truth set of requested slots R_t and predicted set $\hat{R}_t$. Precision, recall, and F1 are computed turn-wise and averaged:

$$\mathrm{RS{-}F1} = \frac{1}{T}\sum_{t=1}^{T}\frac{2\,|\hat{R}_t \cap R_t|}{|\hat{R}_t| + |R_t|}.$$

Graph models often raise RS-F1 by capturing semantic cues (e.g. “Could you tell me the hotel’s address?”) with edges linking requestable slots to their domains.

Active Intent Accuracy (AIA) (Rastogi et al., 2020). For service-rich datasets such as SGD, let i_t and $\hat{i}_t$ be the gold and predicted active user intent for turn t:

$$\mathrm{AIA} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{1}\big[\hat{i}_t = i_t\big].$$

Average Goal Accuracy (AGA). SGD also reports a softer goal metric that averages slot-level correctness *within* each turn:

$$\mathrm{AGA} = \frac{1}{T}\sum_{t=1}^{T}\frac{1}{|s_t|}\sum_{(s,v)\in s_t}\mathbf{1}\big[\hat{v}_{t,s} = v\big].$$

Graph-augmented trackers (e.g. SST, DSTQA) consistently record higher AGA scores, underscoring their robustness to unseen slots and values.

As Table 1 indicates, incorporating graph structure (models marked †) leads to higher joint accuracy. For example, SST’s schema graph helped it reach about 68–70% JGA on MultiWOZ 2.4 (up from 51% on 2.1), comparable to the best models of its time. The copy-augmented GCDST and the triple-copy model TripPy likewise outperform earlier approaches lacking relational mechanisms. Improvements between MultiWOZ 2.1 and 2.4 are substantial – e.g. DST-STAR jumps from 56.4% to 73.6% JGA – reflecting both cleaner data and better exploitation of graph reasoning. On MultiWOZ 2.4, top approaches achieve JGA in the 70–75% range and slot-level accuracy above 97%.

4.3 Graph versus non-graph behaviour

Graph inductive bias directly addresses two longstanding DST pain points: *coreference resolution* (linking repeated or synonymous mentions) and *state carry-over* (propagating values across distant turns). Message-passing layers in DSTQA or DSGFNet propagate embeddings so that, for example, a “restaurant-name” node can influence “taxi-destination” when the user says “take me there.” Non-graph models such as vanilla BERT rely on fixed-length context windows and therefore degrade on longer dialogues.

The benefits remain clear even with large language models (LLMs): GPT2-GAT matches or exceeds a purely generative GPT-2, indicating that symbolic structure still complements distributional knowledge. Moreover, hybrid paradigms – wherein an LLM produces candidate states that are *verified* or *re-ranked* with graph neural networks – show promise for scaling to open-domain ontologies.

4.4 Future research directions

- **Continual ontology expansion.** Real-world assistants must ingest new domains on the fly. Dynamic graph growth, potentially coupled with meta-learning, could allow trackers to append new nodes and edges without catastrophic forgetting.

- **Cross-lingual graph alignment.** Multilingual KGs that link surface forms across languages could accelerate zero-shot transfer, reducing reliance on translation pipelines.

- **Multimodal grounding.** Extending graphs to include visual or haptic entities would unify state tracking across speech, vision, and touch – vital for embodied agents.

- **Explainability dashboards.** By surfacing attention weights or traversal paths through the

Table 1: State-of-the-art dialogue-state trackers on the **MultiWOZ 2.4** test split (results reported in the original papers or their public re-runs as of Aug 2025). † = model uses an explicit graph or graph-augmentation module.

Model	**Graph?**	**JGA (%)**
BREAK-T5 (Won et al., 2023)	–	90.7
MetaASSIST (Ye et al., 2022)	–	80.1
HFSG-DST† (Liao et al., 2025)	+	78.1
Noetic-Graph DST† (Li et al., 2025)	+	78.0
HS2DG-DST† (Gong et al., 2025)	+	77.0
ECDG-DST† (Zhu and Xu, 2025)	+	76.1
MoNET (Zhang et al., 2022)	–	76.0
D3ST-*XXL* (Zhao et al., 2022)	–	75.9
STAR (Ye et al., 2021b)	–	73.6
DSGFNet† (Feng et al., 2022)	+	72.9
SST† (Chen et al., 2020)	+	68.2
SOM-DST (Kim et al., 2019b)	–	66.8
GPT2-GAT† (Lin et al., 2021b)	+	65.3
TripPy† (Heck et al., 2020)	+	64.8
SUMBT (Lee et al., 2019)	–	61.9
DSTQA† (Zhou and Small, 2019b)	+	60.5
TRADE (Wu et al., 2019)	–	55.1

belief graph, practitioners can debug errors and satisfy regulatory transparency requirements.

5 Limitations

Despite their successes, graph-based DST models entail several challenges:

1. **Graph construction overhead.** Building and updating per-turn graphs (especially dynamic ones) incurs latency unsuitable for low-resource or on-device deployments. Approximate or batched updates may be required.

2. **Incomplete or biased relations.** Human-curated schema graphs might omit subtle correlations (e.g. allergy-related restaurant slots) or embed cultural bias. Learned graphs mitigate this but demand large, diverse corpora.

3. **Scalability to open ontologies.** When the slot space explodes (e.g. thousands of product attributes), dense message passing becomes $\mathcal{O}(|\mathcal{S}|^2)$. Sparse or hierarchical graph formulations remain an open problem.

4. **Limited multimodal maturity.** Current multimodal graph approaches often rely on synthetic datasets; real-world noise (blurry images, ASR errors) can break graph alignment.

5. **Evaluation granularity.** Metrics such as JGA are unforgiving yet still miss pragmatic nuances (e.g. partial value matches). Future work should blend symbolic correctness with user satisfaction signals.

5.1 Conclusion

Knowledge graphs have advanced the state-of-the-art in dialogue state tracking by embedding structured reasoning into the tracking pipeline. Empirical evidence across benchmarks such as MultiWOZ and SGD shows consistent gains in joint goal accuracy, especially in complex multi-domain or zero-shot settings. As dialogue systems grow to encompass multiple modalities and languages, the representational flexibility and inductive biases of graphs will likely become even more central to robust DST. The methods surveyed here provide a solid foundation for the next generation of intelligent, knowledge-aware dialogue agents.

6 ConvoTree: A Neural Consolidation Engine for Long-Form Dialogue

To demonstrate the practical application of our taxonomic principles, we introduce ConvoTree, a proof-of-concept system that explores approaches to managing conversation memory through knowledge graphs. ConvoTree serves as an experimental platform for investigating how graph-enhanced DST techniques can be combined, providing insights for future system development.

6.1 System Overview

ConvoTree implements a novel approach to maintaining conversational coherence across extended interactions by combining three key innovations:

Dynamic Graph Construction: Following the "Dynamic Turn-Level Graphs" pattern from our taxonomy, ConvoTree constructs and updates knowledge graphs incrementally as conversations unfold. Each dialogue turn triggers extraction of RDF triples $(subject, predicate, object)$ that capture salient facts and relationships, which are then integrated into the evolving conversation graph using entity matching heuristics.

LLM-Powered Consolidation: ConvoTree implements a simplified "dreaming" process that uses large language models to identify relationships between entities in the knowledge graph. This consolidation mechanism explores how LLM reasoning can be applied to discover implicit connections and merge related concepts, providing a foundation for more sophisticated memory management approaches.

Knowledge-Augmented Interaction: The system explores "Graph-Guided Generation" by maintaining a persistent knowledge graph alongside conversation history. The command-line interface allows users to query and manipulate the knowledge graph, providing a testbed for experimenting with graph-augmented dialogue approaches.

6.2 Prototype Capabilities

ConvoTree demonstrates several key capabilities as a proof-of-concept system:

- **Persistent Knowledge Storage:** SQLite-based storage maintains conversation knowledge across sessions, allowing users to resume conversations with preserved context.
- **LLM-Powered Reasoning:** Integration with OpenAI GPT models enables the system to discover implicit relationships between entities and consolidate related concepts.
- **Interactive Exploration:** Command-line interface provides direct access to knowledge graph operations, supporting experimentation with different consolidation strategies.

While comprehensive performance benchmarking remains future work, initial experiments suggest promising potential for memory-efficient conversation management through knowledge graph consolidation.

6.3 Taxonomic Integration

ConvoTree explores how taxonomic dimensions can be combined in practice: (1) dynamic graph construction through entity extraction, (2) LLM integration for relationship discovery, and (3) domain-agnostic operation through general-purpose knowledge representation. While the current prototype focuses on core functionality rather than optimization, it provides valuable insights into the challenges and opportunities of implementing graph-enhanced dialogue systems.

6.4 Open Source Impact

ConvoTree is released as an open-source project[1] to support research in graph-enhanced dialogue systems. The modular architecture and clear documentation facilitate experimentation with different approaches to knowledge graph construction and consolidation. Potential applications include customer service, education, and creative writing systems that require persistent memory across conversations.

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A ConvoTree Implementation Details

This appendix provides implementation details for the ConvoTree prototype, describing the current system architecture and identifying areas for future development.

A.1 System Architecture

ConvoTree's current implementation consists of several core components:

A.1.1 Real-Time Graph Construction

The system processes each dialogue turn through a knowledge extraction pipeline. When a new message is received through the command-line interface, the system performs the following:

Entity and Relation Extraction: The system uses LLM prompting to identify salient facts from the current turn, formatting them as RDF triples $(subject, predicate, object)$. This approach leverages the language model's understanding to extract entities and relationships.

Graph Integration: Extracted facts are added to the SQLite-backed knowledge graph. The system uses simple string matching and similarity heuristics to identify potential entity matches, though more sophisticated coreference resolution remains future work.

The current implementation adds new triples directly to the database without complex optimization, providing a foundation for future incremental update strategies.

A.1.2 Neural Memory Consolidation

ConvoTree implements a simplified "dreaming" process that consolidates conversation memory through LLM-powered reasoning:

Current Implementation: The system uses OpenAI GPT models to analyze the knowledge graph and identify potential relationships between entities. When the `/dream` command is invoked, the system: - Retrieves entities from the knowledge graph - Identifies similar entities through embedding similarity - Prompts the LLM to discover implicit relationships - Updates the graph with newly discovered connections

Future Enhancements: The current implementation provides a foundation for more sophisticated consolidation approaches, including temporal abstraction, progressive compression, and hierarchical memory organization.

A.1.3 LLM-Powered Reasoning System

The current implementation provides basic knowledge-augmented interaction:

Knowledge Retrieval: When processing user queries through the CLI, the system can access the stored knowledge graph to retrieve relevant facts and relationships.

Context Integration: Retrieved knowledge is combined with conversation history to maintain context across sessions. The current implementation uses simple concatenation, with more sophisticated integration methods planned for future versions.

Response Generation: The system uses standard OpenAI API calls with retrieved context included in the prompt. Advanced techniques for graph-aware generation remain areas for future research.

A.1.4 Long-Term Context Management

The current implementation uses a simplified persistence model:

SQLite Storage: All conversation data and knowledge graph triples are stored in a SQLite database with equal priority. The system does not currently implement temporal hierarchies or progressive compression.

Future Work: Multi-resolution memory hierarchies, temporal abstraction, and adaptive compression strategies represent important areas for future development to enable truly scalable long-term conversation management.

A.2 Implementation Specifications

A.2.1 Command-Line Interface

ConvoTree provides a CLI for interaction and experimentation:

- `/dream`: Triggers the LLM-powered consolidation process to discover relationships
- `/knowledge`: Displays the current knowledge graph contents
- `/reason`: Performs reasoning over the knowledge graph
- `/compress`: Initiates knowledge compression (simplified implementation)
- `/help`: Shows available commands and usage information

Future API Development: A RESTful API with endpoints for programmatic access remains an important goal for production deployment.

A.2.2 Data Structures

The system employs several key data structures:

Conversation Graph: Implemented as a directed multigraph where nodes represent entities (people, places, concepts) and edges represent relationships with associated confidence scores and temporal metadata.

Memory Hierarchy: Structured as a time-based index with pointers to different resolution levels. Each level maintains appropriate data structures (full transcripts, compressed summaries, or learned embeddings).

Consolidation Queue: Priority queue for managing background consolidation tasks, with priorities based on conversation importance, recency, and available computational resources.

A.2.3 Performance Optimizations

The current implementation includes basic optimizations:

Basic Caching: Simple caching mechanisms are implemented for knowledge graph access to reduce database queries.

Future Optimizations: Production deployment would require implementing incremental graph processing, asynchronous consolidation, sophisticated caching strategies, and adaptive resource management. These represent important areas for future development.

A.3 Evaluation Framework

A.3.1 Datasets

A comprehensive evaluation framework has been designed but not yet implemented:

Proposed Datasets: Future evaluation would utilize extended MultiWOZ conversations, customer service logs, and synthetic long conversations to test memory consolidation effectiveness.

Current Status: The system currently undergoes functional testing only. Systematic performance evaluation, benchmarking, and comparison with baseline systems remain important future work.

A.3.2 Metrics

Proposed evaluation methodology would focus on:

Memory Management: Measuring knowledge retention, consolidation effectiveness, and storage efficiency across different conversation types and lengths.

System Performance: Assessing response quality, consistency maintenance, and computational efficiency for extended dialogues.

Implementation Priority: Developing robust evaluation metrics and benchmarks represents a critical next step for validating the system's effectiveness.

B Future Work

The ConvoTree prototype provides a foundation for several important research directions:

B.1 System Development

RESTful API: Developing production-ready API endpoints for programmatic access, including real-time processing, batch operations, and system monitoring capabilities.

Performance Optimization: Implementing incremental graph processing, asynchronous consolidation, sophisticated caching strategies, and adaptive resource management for scalable deployment.

Three-Phase Consolidation: Developing the biologically-inspired replay-consolidation-integration cycle described in our design, including specialized models for each phase.

B.2 Evaluation and Benchmarking

Comprehensive Metrics: Implementing systematic evaluation frameworks for measuring memory efficiency, response quality, and computational performance across different conversation types and lengths.

Benchmark Datasets: Creating standardized datasets for long-form conversation evaluation, including extended MultiWOZ conversations, customer service logs, and synthetic conversations with known ground truth.

Comparative Studies: Conducting rigorous comparisons with baseline systems to quantify the benefits of graph-enhanced memory consolidation.

B.3 Advanced Features

Temporal Memory Hierarchies: Implementing multi-resolution memory management with progressive abstraction from short-term to long-term storage.

Federated Learning: Extending consolidation across multiple conversation instances while preserving privacy through differential privacy techniques.

Multimodal Integration: Supporting visual and audio inputs through specialized graph node types and cross-modal attention mechanisms.

Explainability: Enhanced visualization and explanation capabilities for understanding consolidation decisions and graph evolution patterns.